

StudyRoad Academic Excellence Series

Summary Note: A-Level Biology (High School Grades 10-12)

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Chapter 1: Cell Biology & Transport Mechanisms

- Cell Organelles:
 - Nucleus: Controls genetic material (DNA/RNA) and transcription.
 - Mitochondria: Double membrane, ATP production via Krebs cycle & ETC.
 - Chloroplast: Thylakoid (Light reaction) & Stroma (Calvin cycle).
 - Endoplasmic Reticulum: RER (Protein synthesis) vs SER (Lipid synthesis & detoxification).
- Membrane Transport:
 - Passive Transport: Simple Diffusion, Facilitated Diffusion (Carrier/Channel proteins), Osmosis.
 - Active Transport: Requires ATP against concentration gradient (e.g., Na⁺/K⁺ pump).
 - Bulk Transport: Endocytosis (Phagocytosis/Pinocytosis) and Exocytosis.
- Cell Division:
 - Mitosis: $2n \rightarrow 2n$ (Somatic growth, genetically identical).
 - Meiosis: $2n \rightarrow 1n$ (Gamete production, Genetic variation via Crossing Over in Prophase I).

Chapter 2: Bioenergetics & Respiration

- Cellular Respiration (Yield: 30-32 ATP per Glucose):
 1. Glycolysis (Cytosol): Glucose $\rightarrow$ 2 Pyruvate + 2 ATP + 2 NADH.
 2. Acetyl-CoA Formation (Mitochondrial Matrix): 2 Pyruvate $\rightarrow$ 2 Acetyl-CoA + 2 NADH + 2 CO₂.
 3. Krebs Cycle (Matrix): Produces 2 ATP, 6 NADH, 2 FADH₂, 4 CO₂.
 4. Electron Transport Chain (Inner Membrane/Cristae): Oxidative phosphorylation yielding most ATP.
- Photosynthesis:
 - Light Dependent Reaction: Occurs in Thylakoid membrane, uses H₂O & Light, yields ATP, NADPH, and O₂ byproduct.
 - Calvin Cycle (Light Independent): Occurs in Stroma, fixes CO₂ using RuBisCO enzyme to produce G3P / Glucose.

Chapter 3: Genetics & Biotechnology

- Mendel's Laws:
 - Law of Segregation: Alleles separate during gamete formation.
 - Law of Independent Assortment: Genes on different chromosomes assort independently (9:3:3:1 Dihybrid ratio).
- Molecular Biology Central Dogma:
 - DNA Replication: Semi-conservative, DNA Polymerase (5' $\rightarrow$ 3' direction).
 - Transcription: RNA Polymerase synthesizes mRNA from DNA template.
 - Translation: Ribosome reads mRNA codons, tRNA brings corresponding amino acids.
- DNA Technology (High Exam Frequency):
 - PCR (Polymerase Chain Reaction): Denaturation (95°C) $\rightarrow$ Annealing (55°C) $\rightarrow$ Extension (72°C).
 - Restriction Enzymes: Cut DNA at specific palindromic sequences.

Chapter 4: Human Physiology Key Highlights

- Digestive System: Salivary Amylase $\rightarrow$ Pepsin (Stomach, Acidic pH 1.5-2.0) $\rightarrow$ Trypsin & Lipase (Pancreas/Small Intestine).
- Nervous System: Action Potential (Depolarization via Na⁺ influx, Repolarization via K⁺ efflux).

- Immune System: Innate Immunity vs Adaptive Immunity (B-cells produce Antibodies; T-cells direct cellular attack).